Shubham Jadhav | CS24S009

Department of Computer Science & Engineering, IIT Madras

Portfolio: shubham-jadhav-FiatLux.github.io
GitHub: github.com/shubham-jadhav-FiatLux

Education

Year	Degree / Certificate	Institute	CGPA / %
2024 – Present	M.S. by Research / CSE	Indian Institute of Technology Madras	7.78 / 10
2017 – 2021	B.Tech / CSE	Savitribai Phule Pune University	9.08 / 10
2017	Class XII (Maharashtra State Board)	Vivekanand College Kolhapur	82.63 %
2015	Class X (Maharashtra State Board)	Gadhinglaj High School Gadhinglaj	94.20 %

Technical Skills

Languages	C and C++, Python, SQL
HPC	CUDA, OpenMP, OpenACC, MPI, HIP
Rendering	OpenGL, WebGL, DirectX, Three.js, GLSL, Win32 SDK
Tools	Linux/POSIX, LLVM, Docker, Ghidra, Git/GitHub, LaTeX, Slurm, LLM, Ollama, LangChain
Course Work	**PG Courses:** GPU Programming, Parallel Scientific Computing, Secure Systems Engineering, Program Analysis, Advanced Data Structures and Algorithms, Advanced Computer Architecture **UG Courses:** Operating Systems, Data Structures and Algorithms, Computer Architecture, Database Management Systems, Computer Networks, Compiler Design

Research

Capacitated Vehicle Routing Problem Jul'24 – Present
M.S. Thesis | Guide: Prof. Rupesh Nasre, Prof. Narayanaswamy N S

Context & Problem	The Capacitated Vehicle Routing Problem (CVRP) is an **NP-hard combinatorial optimization problem** which asks for a set of **minimum-cost routes** that start and end at a single depot. A homogeneous fleet of vehicles must service a set of customers with known demands, visiting each exactly once without exceeding vehicle capacity constraints. CVRP underpins modern logistics and distribution planning, where marginal savings in traveled distance yield massive **reductions in cost and emissions**.
Approach & Contributions	• **Engineered a high-performance C++ solver** utilizing a novel geometry-aware construction, highly granular local search, and multi-threaded ruin-and-recreate metaheuristics. • **Pioneered an Adaptive Spiral Recombination architecture** to dynamically decompose and merge sub-problems, enabling the engine to scale efficiently on massive datasets where traditional routing algorithms asymptotically fail. • Discovered **new Best-Known Solutions (BKS) on million-scale instances**, achieving a **10× runtime reduction** compared to SOTA. • **Innovated a Geometry-Aware Initialization** that outperforms existing construction methods in both **speed and scalability**.

Projects

TaintGate: Information-Flow Control for LLM Agents | *Python, MCP, LangChain, AgentDojo* Jan'26 – May'26
Secure Systems Engineering + Program Analysis + LLMs

- Engineered a deterministic **reference monitor** and **MCP proxy** to secure LLM agents against **prompt injection**, replacing brittle probabilistic guardrails.
- Enforced strict **information-flow control** via a **Biba** × **Bell-LaPadula** lattice, utilizing session-level **taint propagation** and backward **program slicing** to dynamically generate source-to-sink explanations for blocked actions.
- Evaluated defense efficacy via the **AgentDojo** benchmark (629 vectors), plotting precision-utility frontiers to quantify mitigation rates across models.

MiniDLC: End-to-End Deep Learning Compiler | *C++, Python, LLVM, CUDA* Aug'25 – Dec'25
GPU Programming + Program Analysis + Computer Architecture

- Engineered an end-to-end compiler lowering models from a custom **Python frontend** into native machine code via C++ **SSA Graph** and **Loop IRs**.
- Implemented **reverse-mode autodiff** as a graph pass, optimized cache locality via dependence-checked **operator fusion**, and engineered liveness-based **memory planning** using interference coloring.
- Generated vectorized CPU code via **LLVM ORC JIT** and tiled GPU kernels via **CUDA NVRTC**, successfully training an **MNIST MLP** and **Char-GPT**.

Valley of Peace: Interactive 3D Portfolio | *Three.js, GLSL, Web Audio* Sep'21 – May'24
Portfolio: shubham-jadhav-FiatLux.github.io

- Engineered an explorable, fully **procedural** 3D world (**21k LoC**) with **zero model, texture or audio files**, all generated in code.
- Rendered **1.7M triangles in ∼200 draw calls** via **GPU instancing** & mesh batching, with **adaptive quality** for low-end devices.
- Directed a **5-min cinematic walkthrough** mode using an autopilot and a camera director on arc-length **Bézier splines**.

Other Projects

- **Coursework Implementations :** cache, branch-predictor, Tomasulo out-of-order & **MESI** coherence simulators; **big.LITTLE** scheduler; **LLVM** range-analysis, Andersen points-to & array bounds-check passes; **CUDA** Borůvka MST and CPU-vs-GPU sine-wave simulation; stack-canary bypass, **ret2libc/ROP**, format-string & glibc heap exploits; **MPI+OpenMP** ray tracer and parallel Poisson solvers.
- **Real-Time Rendering :** built **260+** rendering programs and **15+** mini-projects across **6 platforms** — **Windows** (Win32 SDK), **Linux** (X11/GLX), **Android** (Java), **macOS** & **iOS** (Objective-C++), **Web** (JavaScript) — covering projections, 3D transforms, texturing, model loading and per-fragment Phong lighting in **OpenGL** fixed-function & programmable pipelines, **OpenGL ES**, **WebGL** and **Direct3D 11**, with **GLSL**/**HLSL** shaders and **CUDA**/**OpenCL**–OpenGL interop. **YouTube :** www.youtube.com/@shubham-jadhav-FiatLux
- **PENETRALIA** – Co-engineered a real-time graphics rendering demo featuring a CUDA-FFT ocean solver, procedural terrain, atmospheric scattering, and an advanced post-processing pipeline (godrays, bloom, depth of field) guided by Bézier camera animation. https://youtu.be/Ze5QbGQnguM
- **Asa He Chanchal Maan** – Demo created using the OpenGL graphics library and C language. Win32 SDK is used for windowing purposes and data structures in C are used to render scenes. https://youtu.be/4pB2rZeWkNU

Positions of Responsibility & Other Activities

MS Representative	Elected representative for **MS by Research scholars** in CSE at IIT Madras; collaborated with CSE faculty through class committee meetings to present student feedback on course delivery, quizzes, and assessments, and discuss course structure and grading.
Teaching Assistant	Served as TA for **Systems** and **Algorithms** courses, including GPU Programming, Blockchain Technology, and Natural Language Processing; designed problem sets, assignments and exams, conducted tutorials and office hours, and mentored students.
Event Coordinator	Coordinated logistics and technical operations for major IIT Madras events, including the **ICPC India Online Round** (2025, 2026) and the CSE Degree Distribution Ceremony.
NSS Head	Headed the **National Service Scheme** unit during B.Tech; directed community initiatives including street plays, blood donation drives, check-dam construction, tree plantation drives, and a 7-day rural development camp in Varoti Bk., Pune.
RSP Leader	Led the **Road Safety Patrol** platoon during high school, overseeing team drills and public traffic safety awareness drives.

Interests & Hobbies

Arts: Drawing, Painting, Calligraphy | **Music:** Tabla, Guitar | **Sports:** Table Tennis, Cricket, Volleyball, Carrom, Chess, Kho-Kho